International Institute of Professional Studies

IC-801B MCA VIII(A and B section)

Test #1 Mobile and Wireless Computing

Q1	Explain multi-path propagation and its effects.	4
Q2	Explain direct sequence spread spectrum.	4
Q3	Explain hidden and expose terminal problem.	4
Q4	Explain demand assignment multiple access with implicit reservation.	4
Q5	Explain concept of frequency reuse and handoff.	4
Q6	Explain steps of packet delivery to and from the mobile node in MobileIP.	3

International Institute of Professional Studies,DAVV

MCA VIII

IC-801B Mobile and Wireless Computing

Test#2

Time:1hr MM:20

Q1	Explain function architecture of GSM.	3
Q2	Explain localization and calling in GSM.	3
Q3	Compare a) Infrared Vs Radio transmission b) Infrastructure Vs ad-hoc networks	3
Q4	Explain architecture of IEEE 803.11	4
Q5	What is Bluetooth? In which scenario it is required? What is Piconet and Scatternet?	4
Q6	Explain steps of packet delivery to and from the mobile node in MobileIP.	3

Q.3 Describe the union, in[illegible] theory. Provide examples for each operation. Given two fuzzy sets A and B, compute the union, intersection, and complement of A. Show all steps.

A={(x1,0.4),(x2,0.7),(x3,0.2)}

B={(x1,0.6),(x2,0.3),(x3,0.8)}

8

.....

International Institute of Professional Studies, DAVV, Indore

Class Test – I

Class: - MCA(2 Yrs.) VIII Sem (Sec A & B)

Subject Name: - Soft Computing

Subject Code: - IC-812A

Max Time: - 60 Mins.

MM:20

Note: - All questions are compulsory.

Q.1 Explain the main components of soft computing and how they differ from traditional hard computing techniques. 6

Q.2 Explain the concept of a membership function in fuzzy logic. Provide an example. 6

Q.3 Describe the union, intersection, and complement operations in fuzzy set theory. Provide examples for each operation. Given two fuzzy sets A and B, compute the union, intersection, and complement of A. Show all steps.

A={(x1,0.4),(x2,0.7),(x3,0.2)}

B={(x1,0.6),(x2,0.3),(x3,0.8)} 8

.....

International Institute of Professional Studies, DAVV, Indore

Class Test – II

Class: - MCA (2 Yrs.) VIII Sem (Sec A & B)

Subject Name: - Soft Computing

Subject Code: - IC-812A

Max Time: - 60 Mins.

MM:20

Note: - All questions are compulsory.

Q.1 Draw and explain the architecture of Fuzzy Inference System. 5

Q.2 Create a Mamdani -style FIS for a washing machine with:

- Inputs: Dirt level, Fabric type
- Output: Wash time (constant values)

5

Q.3 What is an Artificial Neural Network (ANN)? Describe the basic structure of an ANN. 5

Q.4 Differentiate between Single layer feed forward neural networks (FNN) and recurrent neural networks (RNN). 5

.....

	q5	q4
q3	q0	q5
q4	q0	q6

Sub. Code: IC-812

Roll No. _____

International Institute of Professional Studies, DAVV

Internal Exam-1 Date: 20/02/2025 Day: Thursday

MCA (5 Years) VIII Semester Section A & B

Theory of Computation

Time: - 1 hour

Max. Marks 20

Note: All questions are compulsory

1. Write formal definition of grammar. Construct a context free grammar over {a,b}* to generate set of all palindrome strings. (2+2)
2. Define ambiguous grammar. Prove that the grammar with following production rules is ambiguous: - (2+2)

   $S \rightarrow a \mid abSb \mid aAb$ $\quad$ $A \rightarrow bS \mid aAAb$
3. Consider the following grammar:

   $S \rightarrow AA \mid CD \mid bB$ $\quad$ $A \rightarrow aA \mid a$ $\quad$ $B \rightarrow bB \mid bC$ (4)

   $C \rightarrow cB$ $\quad$ $D \rightarrow dD \mid d$

   Construct an equivalent grammar without useless symbols
4. Explain Chomsky Normal Form. Reduce the following grammar to Chomsky Normal Form: (2+2)

   $S \rightarrow aAA \mid bBB$ $\quad$ $A \rightarrow a \mid aA$ $\quad$ $B \rightarrow b \mid bB$
5. Consider the following grammar G with production rules: (2+2)

   $S \rightarrow AB$ $\quad$ $A \rightarrow BB \mid a$ $\quad$ $B \rightarrow AB \mid b$

   Find using CKY algorithm whether the string **aaabab** and **aabbb** are accepted by the above grammar

Roll No. _____ Sub. Code: IC-812

International Institute of Professional Studies, DAVV
Internal Exam-1 Date: 02/04/2025 Day: Wednesday
MCA (5 Years) VIII Semester Section A & B
Theory of Computation

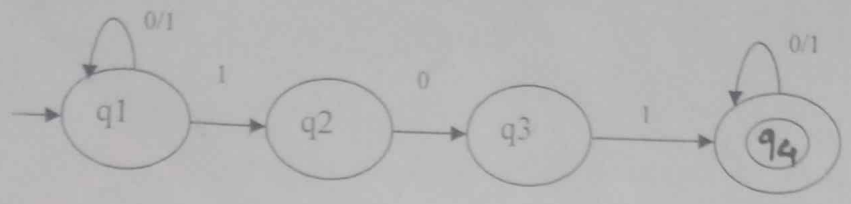
Time: - 1 hour **Max. Marks 20**

Note: **All questions are compulsory**

1. Construct Deterministic Finite Automata for following languages: **(2+2)**

   i) L_1 = {set of all strings over {0,1}* that ends with **110**}
   ii) L_2 = { set of all strings over {0,1}* that ends in **0** and does not contain substring **11**}

2. Write the conditions for automata to be non-deterministic. Convert the following non-deterministic finite automata (NFA) into deterministic finite automata (DFA):- **(3+3)**



3. Construct a minimum state automaton equivalent to the finite automaton described by the following transition table: **(4)**

state	Input	
	a	b
→q0	q0	q3
q1	q2	q5
q2	q3	q4
q3	q0	q5
q4	q0	q6
q5	q1	q4
q6	q1	q3

4. Write the formal definition of Mealy machine and Moore machine. Construct a Moore machine equivalent to the following Mealy machine:- **(3+3)**

Present State	a=0		a=1	
	State	Output	State	Output
→q1	q1	1	q2	0
q2	q4	1	q4	1
q3	q2	1	q3	1
q4	q3	0	q1	1

IC802B/IT805A

International Institute of Professional Studies, DAVV, Indore

MCA (5 Yrs) M. Tech (5 Yrs) VIII Semester

Enterprises Computing Technique

MM 20

Time 1Hrs

1 Explain with Diagram passing by reference in RMI 7

2 How implicit middleware is achieved in EJB 6

3 Write a Session Bean to calculate factorial of a given Number.

Note : write all required interfaces , class and deployment Descriptor 7

Q.3 Explain the association rule-mining problem. Consider an example with the following set of transactions. There are 10 items 5

IC802B

International Institute of Professional Studies, DAVV, Indore

MCA (5 Yrs) VIII Semester

Enterprises Computing Technique

MM 20 Time 1Hrs

1. Differentiate between State full and state less session bean (5)
2. Write notes on Entity bean are view into database. (5)
3. Write an entity Bean with all required interfaces, class and deployment descriptor.(10)

International Institute of Professional Studies

Devi Ahilya University, Indore

MCA-8th Semester Jan-May 2025

First Internal Test

DMW

Note Attempt all questions

Q.1 Suppose that the data-mining task is to cluster the following eight points with (X, Y) representing location into three clusters. **5**

A_1 (4, 6), A_2 (2, 5), A_3 (9, 3), A_4 (6, 9), A_5 (7, 5), A_6 (5, 7), A_7 (2, 2), A_8 (6, 6)

Suppose initially we assign A_1 , A_2 and A_3 as the seeds of three clusters that we wish to apply the K-means method to show:

(a) The three cluster centroids after the first iteration using the Manhattan distance

(b) The final three clusters

(C) Repeat the exercise using the Euclidean distance.

Q2 Using the data in the last exercise, use the first two points as starting seeds of two clusters. Using K-means obtain the clusters and compare the result with the result of Que 1. **5**

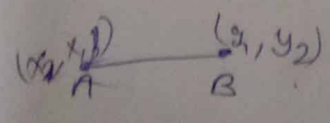
Q.3 Explain the association rule-mining problem. Consider an example with the following set of transactions. There are 10 items. **5**

TID	Items bought
001	B , M , T , Y
002	B , M
003	A , T , S , P
004	A , B , C , D
005	A , B
006	T , Y , E , M
007	A , B , M
008	B , C , D , T , P
009	D , T , S
010	A , B , M

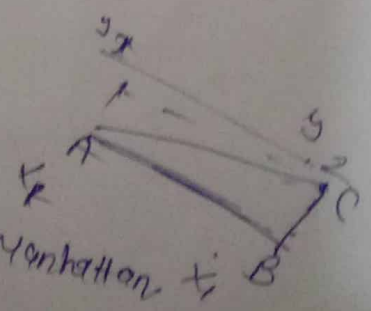
Assume that we wish to find association rules with at least 30% support and 60% confidence. Find the frequent item sets and then the association rules.

Q.4 (a) Give one Suitable example to describe Text Mining System and associated approaches for Selection, Pre-processing ,Transformation , *Data mining* , Interpretation/evaluation **2**

(b) Suggest an application of Text mining and list associated task based on Text Mining that is using algorithms for Anomaly detection, Association rule learning, Clustering , Classification, Regression ,Summarization. **2**



euclidean



Manhattan

Jc-2K21-29

International Institute of Professional Studies
Devi Ahilya University, Indore
M.Tech.(IT)/ MCA -April 2025
Internal Test-2

Q:1 Consider the following table with four items for sale {Bread, Cheese, Juice, Milk, Egg, Yogurt}.

Transaction ID	Items
T100	Bread, Cheese, Egg, Juice
T200	Bread, Cheese, Juice
T300	Bread, Milk, Yogurt
T400	Bread, Juice, Milk
T500	Cheese, Juice, Milk

A. With apriori approach, find association rules with minimum support of 50% and minimum confidence = 75%

B. Apply FP-tree with minimum support of 50% and confidence =75, and demonstrate conditional trees for milk, cheese, and juice.

Q:2 Construct a decision tree to classify bank loan applications into one of three risk classes using the information gain approach. The available data includes the following attributes: **Own Home?** (Y/N),**Married** (Y/N),**Gender** (M/F)**Employed** (Y/N), **Credit Rating** (A, B, C).Predict the **Risk Class** based on these attributes.

Own Home?	Married	Gender	Employed	Credit Rating	Risk Class
Y	Y	M	Y	A	B
N	N	F	Y	A	A
Y	Y	F	Y	B	B
Y	N	M	N	B	C
N	N	F	Y	B	B
N	N	F	Y	A	A
Y	N	M	N	C	C
Y	N	F	Y	B	B
N	Y	F	Y	A	A
N	Y	M	N	B	B
Y	N	F	Y	C	C
N	N	M	N	B	B
Y	N	F	Y	A	A
Y	Y	M	Y	C	C

A. Which attribute would you select as the root node of the decision tree based on information gain?

B. What would the structure of the decision tree look like after applying the information gain approach to the training data?

C. Explain how the decision tree would classify new loan applications based on the attributes provided.

DMW - 3rd internal IC-2K21-29

Q:1 A)What is the difference between OLTP and OLAP in terms of transactions per day?

How many rows will each partition have if 800 million rows are split into 8 partitions?

B)Explain how breaking down large tables into partitions helps performance.

Describe what happens when we roll-up from daily to quarterly data.

C) Calculate the percentage improvement if reporting time goes from 72 hrs to 4 hrs.

If a system processes 500 transactions/sec, how many will it handle in a day?

D) Analyze the network time required to transfer 120 GB at 200 MB/min.

Compare the time taken for 100 OLTP vs OLAP queries.

Q:2 a)What is a star schema?

List the types of OLAP operations.

B) Why is sparsity important in a data cube?

What is the purpose of using a snowflake schema over a star schema?

C) A cube has 3 dimensions: Product (10), Region (5), Time (12). How many cells does it contain?

You slice a cube by Region = "North". How many cells remain?

D) Given 4 dimension tables in a star schema, how many joins are required?

Break down the cost of building 5 departmental data marts, each costing $8000.

Q:3 A)What is support in data mining?

Define classification accuracy.

B)Explain how missing data affects model performance.

Describe the difference between clustering and classification.

C) In a dataset of 500 transactions, {Bread, Butter} appears 120 times. What is the support %?

A classifier gets 180 correct out of 200. What's the accuracy?

D) A KDD process takes 50 hrs, with 15 hrs for data mining. What % is spent on mining?

An algorithm processes 1000 records in 1 sec. Estimate time for 1 million records.

Q:4 A)What is entropy in a decision tree?

What does TF-IDF measure?

B) Explain how term frequency is calculated.

Describe how PageRank uses links to measure importance.

C) Calculate entropy if a dataset has 4 "Yes" and 6 "No" outcomes.

A web page has 800 words; "data" appears 40 times. What is the TF?

D) Evaluate information gain after a split using entropies of subsets.

Find TF-IDF if a word appears 5 times in a document of 100 words, and in 10 of 1000 documents.

Q.5 A)What is temporal mining?

Define spatial density.

B) Why use moving averages in time-series analysis?

What insights can you get from tracking sales patterns over months?

C) Calculate support if a pattern appears in 4 out of 12 months.

What is the Euclidean distance between point A(1,2) and B(4,6)?

D) Given 500 points over 10 km², find density.

Sales data = [100,120,140,160,180]. What is the 3-month moving average at month 4?